



AI-infused knowledge and green intellectual capital: pathways to spur accounting performance drawn from RBV-KBV model and sustainability culture

Saeed Awadh Bin-Nashwan^{a,*}, Jackie Zhanbiao Li^{b,c,**}

^a Department of Accounting, College of Commerce and Business Administration, Dhofar University, Oman

^b Department of Accounting, Finance and Economics, Faculty of Business, Curtin University, Malaysia

^c Banan Hospital Affiliated to Chongqing Medical University, Chongqing, China

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ABSTRACT

AI-infused knowledge (AIK) systems have the potential to transform the way firms operate, providing dynamics for optimizing processes, enhancing decision-making, and fostering innovation. This shift can play a vital role in service-oriented contexts, like accounting firms. In this study, we aim to explore how AIK can drive green intellectual capital (GIC) aspects, spurring accounting firms' performance related to environmental and sustainable practices. Drawn from the resource-based view (RBV) and knowledge-based view (KBV), sustainability culture (SUC) is integrated into the model as a moderator variable to provide an in-depth understanding of this matter. The study adopted a quantitative approach, data was collected through an online survey distributed to accounting professionals. The analysis shows interesting outcomes: AIK positively and significantly affected the internal dimensions of GIC, green human capital (GHC) and green structural capital (GSC) within accounting firms, while green rational capital (GRC) had no effect. In turn, these two internal dimensions, GHC and GSC, had a significant impact on accounting performance across various aspects, including efficiency, accuracy, compliance with environmental standards, innovation in reporting, and timeliness. Intriguingly, SUC was found to significantly moderate the association between GIC dimensions and accounting firm performance. The results of this research offer actionable insights for stakeholders and decision-makers, including accounting firms, clients, and the community at large. Accounting firms are highly recommended to adopt AI-infused platforms to foster GIC dynamics, specifically GHC and GSC. AI-infused knowledge tools have the potential to improve efficiency, accuracy, innovation in reporting, compliance with sustainability standards and timeliness, thus leading to better overall performance. Although service-oriented contexts, like accounting, have the potential to substantially benefit from GIC practices, research in such a significant discipline has been remarkably scarce.

1. Introduction

In today's rapidly evolving business landscape, artificial intelligence (AI) technologies are transforming the way organizations operate, offering tools for optimizing processes, enhancing decision-making, and fostering innovation (Enholm et al., 2022; Loureiro et al., 2021). This shift has significant implications for the accounting industry (Han et al., 2023; Norzelan et al., 2024), where AI-infused knowledge (AIK) systems are enabling firms to streamline tasks, improve accuracy in reporting, and enhance overall performance (Anh et al., 2024). These systems, which encompass advanced data processing, automated analytics, and

decision-support capabilities, are seen as critical assets in maintaining competitive advantage (Ahmad et al., 2023; Norzelan et al., 2024). However, the broader implications of integrating AI into intellectual capital (IC), particularly green IC (GIC), remain an underexplored area of research.

Firms across various industries, including accounting firms, have been forced to rethink their business strategies in recent years due to the growing importance of sustainability in the corporate world (Hsiao et al., 2022; Rodrigue et al., 2023). Regulations and the general movement toward environmental responsibility are putting increasing pressure on firms, particularly in the accounting industry, to adopt

* Corresponding author. Department of Accounting, College of Commerce & Business Administration Dhofar University, 211, Oman.

** Corresponding author. Department of Accounting, Finance and Economics, Faculty of Business, Curtin University, Malaysia

E-mail addresses: s.nashwan233@gmail.com (S.A. Bin-Nashwan), lizhanbiao123@postgrad.curtin.edu.my (J.Z. Li).